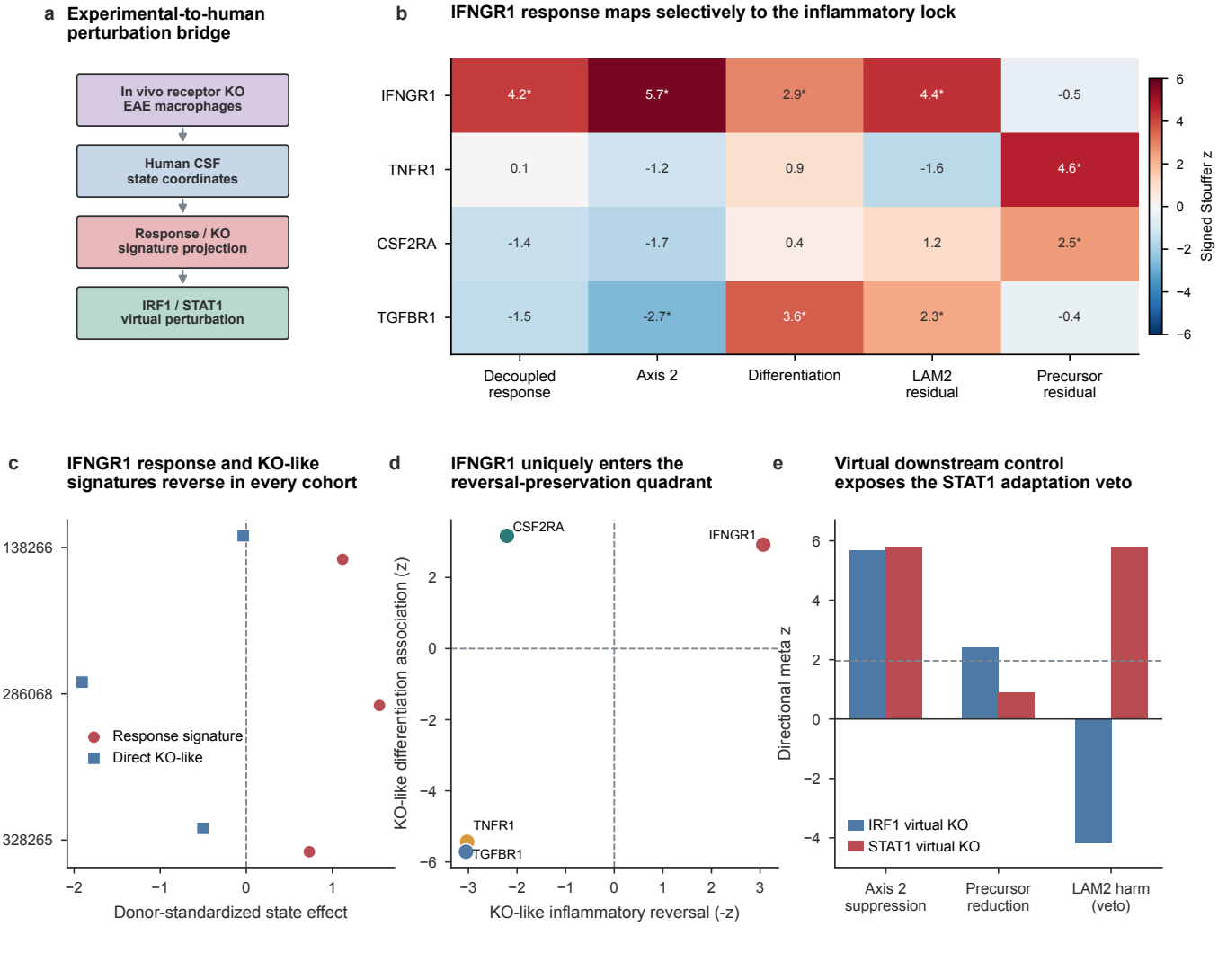

a
Experimental-to-human perturbation bridge
In vivo receptor KO EAE macrophages
Human CSF state coordinates
Response / KO signature projection
IRF1 / STAT1 virtual perturbation
b
IFNGR1 response maps selectively to the inflammatory lock
IFNGR1
TNFR1
CSF2RA
TGFBR1
4.2*
5.7*
2.9*
4.4*
-0.5
0.1
-1.2
0.9
-1.6
4.6*
-1.4
-1.7
0.4
1.2
2.5*
-1.5
-2.7*
3.6*
2.3*
-0.4
Decoupled response
Axis 2
Differentiation
LAM2 residual
Precursor residual
Signed Stouffer z
6
4
2
0
−2
−4
−6
c
IFNGR1 response and KO-like signatures reverse in every cohort
138266
286068
328265
Response signature
Direct KO-like
−2
−1
0
1
Donor-standardized state effect
d
IFNGR1 uniquely enters the reversal-preservation quadrant
CSF2RA
IFNGR1
TNFR1
TGFBR1
KO-like differentiation association (z)
KO-like inflammatory reversal (-z)
2
0
−2
−4
−6
−3
−2
−1
0
1
2
3
e
Virtual downstream control exposes the STAT1 adaptation veto
Directional meta z
6
4
2
0
−2
−4
IRF1 virtual KO
STAT1 virtual KO
Axis 2 suppression
Precursor reduction
LAM2 harm (veto)